Multi-Modal Contactless Battery Sensing Using Commercial Handheld Smartphones

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ABSTRACT

Lithium-ion battery state estimation traditionally relies on electrical measurements that degrade over time and require costly hardware. We present CoBas (Contactless Battery Sensing), a multimodal pipeline for contactless state of charge (SoC) classification using only a commercial smartphone. CoBas emits near-ultrasonic chirps from a laptop speaker, captures the acoustic and visual response on an iPhone, and fuses STFT spectral features with synchronized video frames using a dual-stream ResNet-18 with late fusion. Evaluated on Samsung 21700 lithium-ion cells at 0%, 50%, and 100% SoC, the fused model achieves 100% classification accuracy on held-out test videos. These results provide preliminary evidence that near-ultrasonic frequencies carry discriminative information upon interaction with Li-ion cells, establishing contactless battery diagnostics as a feasible pathway for low-cost smart energy monitoring.

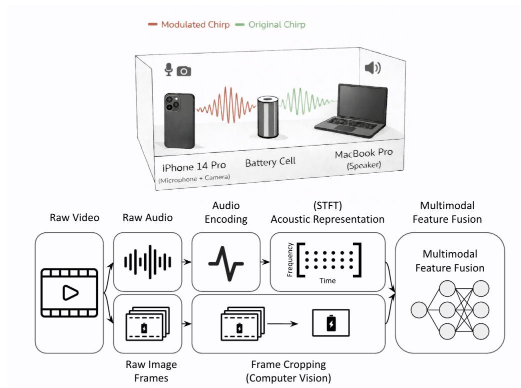


Figure 1: *CoBas* system overview.

KEYWORDS

State of charge (SoC), Lithium-ion batteries, Feature Fusion, Multimodal Learning

ACM Reference Format:

Selase Eric Doku, Pedro Callado de Paiva, Nitin Shankar Madhu, Ningyue Mao, Zheng Liu, Xuan Zhou, Xiao Zhang . 2026. Multi-Modal Contactless Battery Sensing Using Commercial Handheld Smartphones. In *Proceedings of e-Energy '26.* ACM, New York, NY, USA, 2 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

Lithium-ion batteries power everything from consumer electronics to electric vehicles, making reliable state estimation critical for safe and efficient operation. Traditional approaches rely on electrical measurements such as terminal voltage and current integration. Model-driven methods like Coulomb counting and Kalman filters perform well under stable conditions but degrade under dynamic loading and long-term cycling [1, 2]. Data-driven methods using electrical features have gained traction [3], but share a fundamental limitation: they cannot observe internal structural changes as the cell ages, causing feature drift as state of health declines [4].

Ultrasonic sensing offers a physically grounded alternative. Electrochemical processes during lithiation and delithiation induce measurable mechanical and structural changes within the cell, which manifest as variations in acoustic wave propagation [5]. Prior work has shown that guided ultrasonic waves correlate with SoC through changes in wave velocity, attenuation, and dispersion tied to electrode density shifts [6, 7]. Frequency-domain features are particularly well-suited for this task, as they are more resilient to sensor coupling shifts from thermal expansion or mechanical vibration than time-domain features like Time of Flight [8]. However, existing ultrasonic systems depend on high-fidelity oscilloscopes and physical contact with the cell, making them neither portable nor practical for embedded or consumer applications [9].

Air-coupled ultrasonic sensing removes the contact requirement, and recent work has demonstrated its feasibility for SoC estimation [10, 11]. We build on this direction with *CoBas* (Contactless Battery Sensing), a multimodal pipeline for contactless battery SoC classification using only a commercial smartphone. CoBas emits near-ultrasonic chirps from a laptop speaker, captures the response on an iPhone microphone, and fuses STFT spectral features with synchronized video frames using a dual-stream ResNet-18 with late fusion. The result is a low-cost, contactless SoC monitoring system that requires no specialized hardware, no physical contact, and no dispersion modeling.

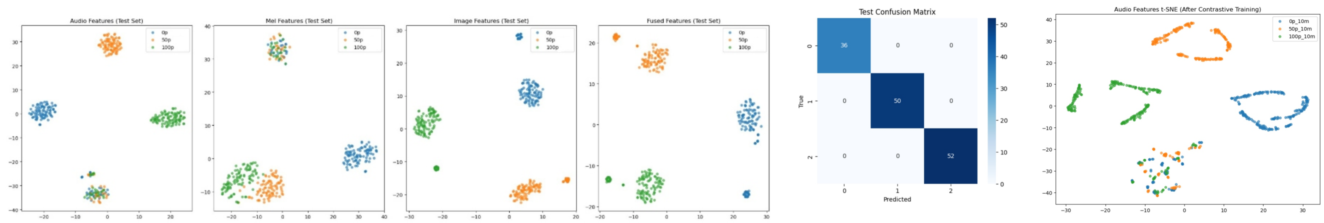
2 SYSTEM OVERVIEW

CoBas classifies lithium-ion cell SoC by fusing near-ultrasonic acoustic responses with synchronized optical signals, captured entirely using commodity smartphone hardware without physical contact with the cell. A MacBook Pro (M1, 2020) emits controlled

e-Energy '26, TBA, TBA, TBA

ACM ISBN 978-x-xxxx-xxxx-x/YY/MM…$15.00

https://doi.org/10.1145/nnnnnnn.nnnnnnn



(a) supervised fused representation [audio, Mel, vision, fused features] (b) SoC confusion matrix (c) unsupervised STFT embedding

Figure 2: Evaluation: (a) supervised fused representation, (b) SoC confusion matrix, and (c) unsupervised STFT embeddings.

chirp signals spanning 15–19.2 kHz, and an iPhone 14 Pro simultaneously captures acoustic and visual data using its built-in microphone and camera. Samsung 21700 cylindrical lithium-ion cells are tested at 0%, 50%, and 100% SoC under identical excitation and environmental conditions.

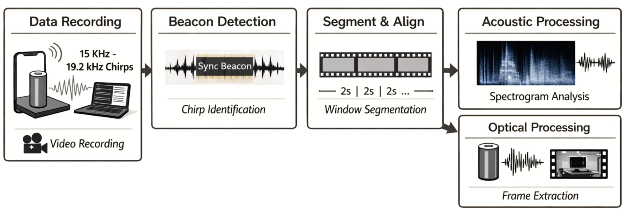


Figure 3: CoBas pipeline: chirp emission, beacon detection, window segmentation, and parallel acoustic and optical processing streams.

As shown in Figure 3, a beacon encoding protocol embedded in each chirp enables automatic boundary detection, eliminating timing variability and achieving a recording length standard deviation of 0.003 ± 0.002 s across trials. Each recording is segmented into aligned 2-second windows, with every audio segment paired to its corresponding video frame. STFT representations are computed and cropped to the 15–19.2 kHz band, preserving linear frequency resolution critical for detecting fine-grained spectral shifts [8].

CoBas processes each modality through an independent ResNet-18 stream before combining embeddings at a late fusion layer for joint classification. The acoustic stream accepts single-channel STFT spectrograms and the optical stream processes synchronized video frames. Late fusion is deliberate: acoustic features encode internal electrochemical state while visual features stabilize the representation against setup variability such as fixture alignment and sensor placement [11].

3 RESULTS

CoBas is evaluated on Samsung 21700 cylindrical lithium-ion cells at 0%, 50%, and 100% SoC under controlled laboratory conditions. The dataset is split 70/20/10 across training, validation, and held-out test videos, with SoC treated as a three-class classification problem.

The fused audio-visual model achieves 100% accuracy on held-out test videos across all three charge states, as shown in Figure 2(b). Audio-only STFT features show partial class separation, visible in Figure 2(a), confirming that near-ultrasonic spectral content carries discriminative information about battery state [8]. Mel spectrogram features show increased class overlap, as perceptual compression attenuates physically meaningful high-frequency structure. Visual features alone do not encode SoC. When fused with STFT features, visual cues reduce sensitivity to setup variability and yield the most separable embeddings across all tested modalities.

To assess whether battery-related structure emerges without label supervision, unsupervised embeddings are learned from STFT features alone, shown in Figure 2(c). The clusters yield a Silhouette Score of 0.1597, Davies-Bouldin Index of 2.1179, and Adjusted Rand Index of 0.0144, indicating weak but non-random acoustic structure that does not align with SoC labels without supervision. Supervised multimodal fusion is therefore necessary to reliably map spectral patterns to battery state.

These results provide preliminary evidence that contactless SoC classification is feasible using only commodity smartphones. The controlled setup and single cell chemistry are acknowledged limitations. Future work will extend evaluation to diverse battery chemistries, dynamic charge-discharge conditions, and continuous SoC regression, working toward a practical and scalable battery diagnostic tool for smart energy systems.

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